

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	31 January 2025
Team ID	_____
Project Name	Car Rental Request Automation in ServiceNow Using Service Catalog and Workflows
Maximum Marks	4 Marks

Technical Architecture:

The proposed architecture is based on the ServiceNow platform. Employees interact with the Service Catalog through the ServiceNow web interface. The catalog request is stored as a Requested Item (RITM). Flow Designer automates approval, conditional processing, Catalog Task creation, retrieval of catalog variables and email notification. The Transport Team fulfills the approved request.

Architecture Diagram

USER / REQUESTER	APPLICATION / CLOUD	APPROVER
ServiceNow Portal Service Catalog	ServiceNow Platform Service Catalog → Requested Item (RITM) ↓ Flow Designer Ask for Approval → If Approved ↓ Create Catalog Task → Get Catalog Variables ↓ Send Email / Update Status	Manager / Authorized Approver Review request Approve / Reject
TRANSPORT TEAM Receives Catalog Task Uses rental details Completes fulfillment	DATA STORAGE Requested Item (sc_req_item) Approval Records Catalog Task (sc_task) Catalog Variables / Request Data	NOTIFICATION ServiceNow Email Approval / Status Update

Infrastructure: ServiceNow is the cloud platform hosting the application, workflow engine, database records and service interface. No separate local application server is required for the proposed implementation. **External interface:** ServiceNow email notification service. **Machine Learning:** Not applicable to the core project.

Table-1 : Components & Technologies

S.No	Component	Description	Technology
1	User Interface	How the user interacts with the application.	ServiceNow Service Portal / Service Catalog UI, HTML, CSS, JavaScript
2	Application Logic-1	Logic for submitting and managing car rental requests.	ServiceNow Service Catalog
3	Application Logic-2	Logic for approval and conditional processing.	ServiceNow Flow Designer
4	Application Logic-3	Logic for fulfillment task creation and notifications.	Flow Designer Actions: Ask for Approval, Create Catalog Task, Get Catalog Variables, Send Email
5	Database	Request, approval and task data storage.	ServiceNow platform tables: sc_request, sc_req_item, sysapproval_approver, sc_task
6	Cloud Database	Database service used by the cloud platform.	ServiceNow platform-managed database
7	File Storage	File/attachment storage requirements.	ServiceNow attachments / platform storage, if required
8	External API-1	Notification interface for requester updates.	ServiceNow Email / Notification service
9	External API-2	Optional future vehicle availability integration.	External car-rental / fleet API, if integrated
10	Machine Learning Model	Purpose of machine learning model.	Not applicable for the current project
11	Infrastructure (Server / Cloud)	Application deployment on local system / cloud.	ServiceNow Cloud Platform; browser-based access; no separate application server

Technology selection rationale: ServiceNow Service Catalog provides the standardized request interface, while Flow Designer provides the automation layer for approvals, conditions, fulfillment tasks, variable retrieval and notifications. ServiceNow-managed data storage keeps request-related records within the same platform.

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	No separate open-source application framework is required for the core implementation. The project is built on the ServiceNow platform.	ServiceNow platform; JavaScript can be used for supported scripting/customization
2	Security Implementations	Role-based access controls should restrict who can submit, approve and fulfill requests. Catalog and table permissions should protect request data.	ServiceNow Roles / ACLs, authentication, authorization and HTTPS/TLS
3	Scalable Architecture	The solution uses a cloud platform with reusable catalog and flow components. More request types, users and fulfillment teams can be added without redesigning the complete process.	ServiceNow Cloud Platform; Service Catalog; Flow Designer
4	Availability	The application is accessed through the ServiceNow cloud platform, avoiding dependence on a separately managed local application server.	ServiceNow Cloud Platform; browser-based access
5	Performance	Use simple catalog forms, efficient Flow Designer actions and only required data retrieval. Avoid unnecessary flow actions and repeated processing.	ServiceNow Service Catalog; Flow Designer; platform-managed database

References

1. C4 Model — <https://c4model.com/>
2. IBM Cloud Architecture — <https://www.ibm.com/cloud/architecture>
3. AWS Architecture — <https://aws.amazon.com/architecture>

Project Architecture Summary: Employee → Service Catalog → RITM → Flow Designer → Approval → Approved condition → Catalog Task → Transport Team → Email/Status update.